Teaching Proteins to Speak: Speech-Inspired Embeddings for Biological Discovery

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Abstract

Protein representation learning has largely focused on modeling amino-acid sequences as static textual data, often overlooking the temporal and functional organization of protein activity within biological systems. In this work, we propose a novel representation framework that treats protein function as a symbolic temporal sequence, inspired by methodologies from speech representation learning. Rather than modeling proteins solely as sequences of residues, we encode functional roles and pathway interactions into symbolic structures analogous to syllables and phrases. These representations are embedded using speech-inspired transformer encoders to explore latent biological patterns in an unsupervised manner. A small proof-of-concept experiment demonstrates that the proposed encoding yields non-random clustering of biological pathways, suggesting that speech-based embeddings may capture complementary structural information beyond conventional protein language models. This work is intended as an exploratory framework and does not aim to provide clinical predictions, but rather to open new directions for symbolic and temporal modeling in computational biology.

1 Introduction

Advances in protein representation learning have led to powerful sequence-based models capable of capturing structural and evolutionary properties of proteins. Models such as protein language transformers have demonstrated strong performance across multiple downstream tasks. However, these approaches primarily focus on local or global sequence patterns and often neglect the functional and temporal organization of protein activity at the system level.

Biological processes such as signaling cascades, metabolic pathways, and regulatory feedback loops are inherently temporal and symbolic in nature. Proteins do not operate in isolation, but rather participate in ordered functional sequences that collectively determine phenotype. Capturing this “flow of function” remains a challenge for existing representations.

Inspired by speech representation learning, which models temporal structure, robustness to noise, and hierarchical symbolic patterns, we propose a novel conceptual framework in which protein function is represented as a symbolic temporal sequence. In this analogy, proteins act as executable units of a biological syllabus, where pathways form structured narratives rather than static graphs.

2 Related Work

2.1 Protein Language Models

Protein language models treat amino-acid sequences as textual data, learning representations that capture biochemical and evolutionary constraints. While effective, these models are primarily

sequence-centric and do not explicitly encode functional execution or pathway-level semantics.

2.2 Pathway and Network Models

Graph-based approaches model protein interactions and pathways but are often static and struggle to encode temporal ordering or symbolic execution of biological functions.

2.3 Biological Sonification

Previous attempts to map biological sequences to sound or music have largely focused on visualization or artistic interpretation, rather than representation learning or pattern discovery.

3 Conceptual Framework

We introduce a representation paradigm in which protein function is treated as a symbolic syllabus executed over time.

- Genes correspond to instructional units.
- Proteins represent executable functional lessons.
- Pathways form ordered symbolic sequences analogous to sentences.
- Biological phenotype emerges from the execution quality of these sequences.

Rather than modeling raw biochemical data directly, we abstract protein roles into symbolic tokens that capture function, interaction, and order of execution.

4 Methodology

4.1 Symbolic Encoding of Protein Function

Each protein is mapped to one or more symbolic tokens derived from curated annotations such as functional domains, pathway membership, or regulatory roles. For example:

$$\text{DNA repair} \longrightarrow \text{repair}$$
$$\text{cell cycle checkpoint} \longrightarrow \text{pause}$$

Ordered protein interactions within pathways are then concatenated to form symbolic sequences.

4.2 Speech-Inspired Embedding

The resulting symbolic sequences are treated analogously to phoneme or syllable sequences in speech. These sequences are embedded using a transformer-based encoder inspired by speech representation models, focusing on temporal structure rather than acoustic realism. Only the encoder component is used, without audio synthesis.

5 Proof-of-Concept Experiment

5.1 Dataset

A small set of proteins was selected from two distinct biological pathways:

- Cell cycle regulation
- Core metabolic processes

Protein annotations were obtained from publicly available databases.

5.2 Experimental Procedure

1. Encode protein functions into symbolic sequences.
2. Embed sequences using a speech-inspired transformer encoder.
3. Perform unsupervised clustering on the resulting embeddings.
4. Compare clustering structure against a randomized baseline.

5.3 Evaluation

Clustering quality was assessed using silhouette scores and low-dimensional visualization techniques such as PCA or UMAP.

6 Results

The symbolic speech-inspired embeddings demonstrated non-random clustering aligned with biological pathway membership. While separation was moderate, it consistently exceeded random baselines, suggesting that the encoding captures meaningful structural information.

7 Discussion and Limitations

This work does not claim biological ground truth or clinical applicability. Instead, it introduces a representation hypothesis that complements existing sequence-based and graph-based approaches. Limitations include reliance on curated annotations, small-scale validation, and abstraction from biochemical detail.

Future work may explore larger datasets, integration with protein language models, and extension to dynamic expression data.

8 Conclusion

We propose a novel speech-inspired symbolic representation of protein function that emphasizes temporal execution and functional organization. By framing biological activity as symbolic sequences, this work opens new directions for exploratory pattern discovery at the systems level. While preliminary, the results suggest that cross-domain representation learning may offer valuable insights into complex biological organization.

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References